📋 NOA ML Master Specification Document

Predicting Sperm Retrieval Success in Non-Obstructive Azoospermia (NOA) Patients

Complete Evidence-Based Machine Learning Pipeline — Publication-Ready

Version: 1.0
Created: 2026-03-14
Project: PhD/Medical Thesis — NOA Sperm Retrieval Prediction
Target: Peer-reviewed journal publication (high-impact urology/andrology/AI-in-medicine)
Language: All code & reports in English; clinical interpretation also in Farsi

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1. PROJECT OVERVIEW

1.1 Goal

Predict **sperm retrieval success** in patients with **Non-Obstructive Azoospermia (NOA)** undergoing micro-TESE (Testicular Sperm Extraction) surgery, using machine learning with full explainability for clinical decision support.

1.2 Scope

- **Primary Outcome ONLY:** Outcome 1 (first sperm retrieval attempt success/failure)
- Outcomes 2, 3, 4 (salvage mTESE) are **excluded** from this project — reserved for future publications
- Binary classification: Success (1) vs. Failure (0)

1.3 Why Outcome 1 Only?

- Higher quality with focused analysis on one outcome
- Potential for 3-4 separate publications from the same dataset (one per outcome)
- Avoids complexity explosion from multi-outcome modeling
- Cleaner clinical narrative for reviewers

1.4 Target Audience

- **Primary:** Physicians, urologists, andrologists, surgeons
- **Secondary:** Peer reviewers, journal editors
- **Tertiary:** Medical center for clinical deployment (CDSS — future phase)

1.5 Key Principles

1. **Evidence-based**: Every decision backed by scientific references (preferably 2015-2025)
2. **Publication-ready**: All outputs formatted for direct journal submission
3. **Novelty-aware**: Test novel approaches; if they outperform established methods, document as contribution
4. **Double-checked**: Validate after every phase before proceeding
5. **Logged**: Complete decision logger with references for thesis/article writing
6. **Clinically grounded**: Statistical + clinical significance both matter

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2. DATASET SPECIFICATION

2.1 Source File

- **Filename:** `Book_11.17.2025_Dr. Rad_Final ID.xlsb`
- **Location:** `[path] Rad_Final ID.xlsb`
- **Format:** Excel Binary (.xlsb)
- **Sheet:** `Final Sheet`

2.2 Dataset Dimensions

- **Rows:** 2,450 patients
- **Total Columns:** 55
- **Usable Feature Columns:** 36 (after exclusions)

- **Target Column:** `Outcome1` (binary: 0 = Failure, 1 = Success)

2.3 Class Distribution (from original data)

- **Failure (0):** ~62.7% (1,534 samples)
- **Success (1):** ~37.3% (916 samples)
- **Imbalance Ratio:** ~1.67:1

2.4 Complete Column Inventory (55 columns)

Identifier Columns (EXCLUDE from modeling)

#	Column	Action
0	ID	Drop
1	Roy-ID	Drop

Feature Columns (36 features for modeling)

Feature_N	Clinical Variable	Category	Original dtype	% Missing	Unique Values
Feature_1	Age	Demographics	int64	0.0%	58
Feature_2	Pathology-RT	Pathology	object	22.2%	772
Feature_3	Pathology-LT	Pathology	object	49.0%	519
Feature_4	Partner Age	Demographics	float64	8.6%	42
Feature_5	Race	Demographics	object	0.0%	13
Feature_6	Infertile family members	Lifestyle/Environmental	object	95.8%	10
Feature_7	Habits	Lifestyle/Environmental	object	42.2%	30
Feature_8	Height	Anthropometrics	float64	67.2%	45
Feature_9	Body Weight	Anthropometrics	float64	67.2%	84
Feature_10	BMI	Anthropometrics	float64	64.1%	529
Feature_11	Occupation (Toxic Exposure)	Lifestyle/Environmental	object	94.4%	7
Feature_12	Diabetes	Comorbidities	float64	99.1%	1
Feature_13	Hypertension	Comorbidities	float64	99.8%	1
Feature_14	Surgery trauma(s)	Surgical/Urological	object	82.0%	163
Feature_15	Varicocele	Surgical/Urological	object	97.1%	11
Feature_16	T.BX (Testicular Biopsy)	Surgical/Urological	float64	95.6%	1

Feature_N	Clinical Variable	Category	Original dtype	% Missing	Unique Values
Feature_17	Inguinal hernia	Surgical/Urological	object	98.7%	3
Feature_18	Orchiopexy	Surgical/Urological	object	97.7%	3
Feature_19	Testis Size right (Sono)	Testicular Measurements	object	82.8%	408
Feature_20	RT Size (Orchidometer)	Testicular Measurements	object	73.8%	31
Feature_21	RT-XYZ (Sono)	Testicular Measurements	float64	83.0%	386
Feature_22	Testis Size left (Sono)	Testicular Measurements	object	82.7%	408
Feature_23	LT Size (Orchidometer)	Testicular Measurements	object	73.6%	34
Feature_24	LT-XYZ (Sono)	Testicular Measurements	float64	83.0%	381
Feature_25	Sakamoto-RT/mL	Testicular Measurements	float64	82.3%	387
Feature_26	Sakamoto-LT/mL	Testicular Measurements	float64	82.8%	382
Feature_27	Testicular volume_RT (Guess)	Testicular Measurements	float64	62.9%	28
Feature_28	Testicular volume_LT (Guess)	Testicular Measurements	float64	62.7%	26

Feature_N	Clinical Variable	Category	Original dtype	% Missing	Unique Values
Feature_29	Seminal plasma pH	Semen Parameters	float64	67.9%	15
Feature_30	Testosterone levels	Hormonal Profile	object	22.6%	578
Feature_31	LH	Hormonal Profile	object	25.1%	1034
Feature_32	FSH	Hormonal Profile	object	22.2%	1259
Feature_33	Prolactin	Hormonal Profile	object	40.4%	163
Feature_34	E2 (Estradiol)	Hormonal Profile	object	41.6%	103
Feature_35	Karyotype	Genetic Factors	object	62.3%	26
Feature_36	Y chromosome microdeletion (AZFa, AZFb, AZFc)	Genetic Factors	object	62.4%	4

Excluded Columns (all-null or irrelevant)

- **All-null columns (drop):** Torsion of spermatic cord, Cancer Type, Cancer Treatment, Infertility length, TSH, Sex Hormone-Binding Globulin (SHBG), Inhibin B, AMH, Surgery Therapeutic
- **Other outcome columns (drop for Outcome 1 focus):** Successful Sperm Retrieval-1 (text), Successful Sperm Retrieval-2 (text), Outcome2, Successful Sperm Retrieval-3 (text), Outcome3, Successful Sperm Retrieval-4 (text), Outcome4

Feature Categories Summary

Category	Count	Features
Demographics	3	Age, Partner Age, Race
Pathology	2	Pathology-RT, Pathology-LT
Anthropometrics	3	Height, Body Weight, BMI
Lifestyle/Environmental	3	Infertile family members, Habits, Occupation
Comorbidities	2	Diabetes, Hypertension
Surgical/Urological	5	Surgery trauma(s), Varicocele, T.BX, Inguinal hernia, Orchiopexy
Testicular Measurements	10	Testis sizes (sono/orchidometer), RT/LT-XYZ, Sakamoto, Volume estimates
Semen Parameters	1	Seminal plasma pH
Hormonal Profile	5	Testosterone, LH, FSH, Prolactin, E2
Genetic Factors	2	Karyotype, Y chromosome microdeletion

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3. EVIDENCE-BASED DECISIONS WITH REFERENCES

> **CRITICAL PRINCIPLE:** Every methodological decision must cite a scientific reference. Prefer references from 2015-2025. The decision logger must record: (1) what was decided, (2) why, (3) the reference, (4) whether it was the best approach or if a novel approach outperformed it.

3.1 Missing Data Handling

Threshold for Dropping Columns

- **Decision:** Drop columns with >50% missing values
- **Rationale:** EMA (European Medicines Agency, 2010) ICH E9(R1) guideline recommends that variables with excessive missingness (>40-50%) introduce unacceptable imputation uncertainty and should be excluded. TRIPOD (Transparent Reporting of a Multivariable Prediction Model, 2015) also recommends reporting and justifying missingness thresholds.

- **References:**
- EMA ICH E9(R1) Addendum on Estimands (2010/2020)
- Collins GS et al. “TRIPOD Statement” BMJ (2015)
- Sterne JAC et al. “Multiple imputation for missing data in epidemiological studies” BMJ (2009)
- ⚠️ **IMPORTANT:** Before dropping, check if the column has clinical importance. If a column is >50% missing but clinically critical (e.g., Karyotype at 62.3%), discuss with the medical team. Create a **missingness indicator variable** (binary: 1=missing, 0=present) before dropping.

Imputation Strategy

- **Decision:** MissForest (Random Forest-based multiple imputation) as primary method
- **Rationale:** MissForest handles mixed data types (continuous + categorical), captures non-linear relationships, and outperforms MICE/mean/median imputation in clinical datasets.
- **References:**
- Stekhoven DJ & Bühlmann P. “MissForest—non-parametric missing value imputation for mixed-type data” Bioinformatics (2012)
- Waljee AK et al. “Comparison of imputation methods for missing laboratory data in medicine” BMJ Open (2013)
- Shah AD et al. “Comparison of random forest and parametric imputation models” Am J Epidemiol (2014)
- **Fallback:** MICE (Multiple Imputation by Chained Equations) if MissForest fails
- **For categorical variables:** Mode imputation with missingness indicator
- **For numerical variables:** MissForest → MICE → KNN Imputation (in order of preference)

Implementation Notes

- Always create **missingness indicator columns** before imputation (feature: `{column}_missing` = 0/1)
- Perform imputation **ONLY on training set**, then apply to test set
- Report percentage of imputed values per column in final report

3.2 Class Imbalance Handling

Decision: SMOTE + Class Weights (Combined)

- **SMOTE:** Applied to training set only (never test set)
- **Class Weights:** Set in model parameters (`class_weight='balanced'` or computed weights)
- **Rationale:** The combination addresses imbalance at both the data level (SMOTE) and algorithm level (class weights), providing superior performance to either method alone.
- **References:**
- Chawla NV et al. “SMOTE: Synthetic Minority Over-sampling Technique” JAIR (2002)
- Fernández A et al. “Learning from Imbalanced Data Sets” Springer (2018) — Chapter on hybrid approaches
- He H & Garcia EA “Learning from Imbalanced Data” IEEE TKDE (2009)
- **Novelty Check:** After implementing SMOTE+Weights, also test:
- ADASYN (Adaptive Synthetic Sampling)
- SMOTETomek (SMOTE + Tomek links cleaning)
- Borderline-SMOTE
- If any novel combination outperforms SMOTE+Weights → document as a contribution

3.3 Feature Selection Strategy

Decision: Combined Statistical + Technical Approach (3-Stage Pipeline)

1. **Stage 1 — Statistical Filter:** Univariate tests (t-test/Mann-Whitney U for numerical, Chi-square for categorical) with $p < 0.05$ threshold
2. **Stage 2 — Recursive Feature Elimination (RFE):** Using Random Forest/XGBoost as estimator, with cross-validation (RFECV)
3. **Stage 3 — Feature Importance Ranking:** Aggregate importance from multiple models

- **Rationale:** This combined approach satisfies both clinical reviewers (who value p-values and statistical significance) and ML best practices (which prioritize predictive importance). Statistical filtering removes noise; RFE finds optimal subset; importance ranking confirms consensus.
- **References:**
- Guyon I & Elisseeff A “An Introduction to Variable and Feature Selection” JMLR (2003)
- Saeys Y et al. “A review of feature selection techniques in bioinformatics” Bioinformatics (2007)
- Chandrashekar G & Sahin F “A survey on feature selection methods” Computers & Electrical Engineering (2014)
- **CRITICAL:** Do NOT blindly drop features based on importance alone. Some clinically important features (e.g., Karyotype, Y chromosome microdeletion) MUST be included regardless of statistical significance. Flag these as “clinically mandated features.”
- **Also run:** Include ALL features in one model run (no feature selection) to compare — this tests whether feature selection actually helps.

3.4 Outlier Detection & Handling

Decision: IQR Method + Clinical Review

- **Detection:** IQR method (Q1 - 1.5×IQR, Q3 + 1.5×IQR) for numerical features
- **Supplementary:** Z-score method ($|Z| > 3$) for confirmation
- **Handling:** Do NOT automatically remove outliers. Instead:
  1. Flag all detected outliers
  2. Document which variable, value, and detection method
  3. Create an **outlier report** for clinical review by the medical team
  4. Only remove/winsorize after clinical confirmation
  5. If medical team is not available: Winsorize (cap at 1st/99th percentile) rather than delete
- **Rationale:** In clinical data, outliers may represent real biological extremes (e.g., very high FSH in Klinefelter patients). Removing them without clinical input introduces bias.
- **References:**
- Aguinis H, Gottfredson RK, Joo H “Best-Practice Recommendations for Defining, Identifying, and Handling Outliers” Organizational Research Methods (2013)
- Osborne JW & Overbay A “The power of outliers (and why researchers should ALWAYS check for them)” Practical Assessment, Research & Evaluation (2004)
- **Output:** Generate `outlier_report.csv` listing all detected outliers with: patient_index, variable, value, IQR_flag, Z_flag, clinical_action_needed

3.5 Random Seeds for Reproducibility

Decision: Multiple Seeds (5-10 seeds)

- **Seeds to use:** [42, 123, 456, 789, 1024, 2023, 2024, 3141] (8 seeds)
- **Report:** Mean ± standard deviation across all seeds for every metric
- **Rationale:** Single-seed results can be misleading due to data split variance. Multiple seeds provide confidence intervals and demonstrate robustness.
- **References:**
- Bouthillier X et al. "Accounting for Variance in Machine Learning Benchmarks" MLSys (2021)
- Picard RR & Cook RD "Cross-Validation of Regression Models" JASA (1984)
- Henderson P et al. "Deep Reinforcement Learning that Matters" AAAI (2018) — on seed sensitivity
- **Implementation:** For each model, train with all 8 seeds, report mean ± SD of AUC, F1, Accuracy, MCC

3.6 Train/Test Split Strategy

Decision: Both 80/20 AND 70/30 splits

- **Primary:** 80/20 stratified split (standard)
- **Secondary:** 70/30 stratified split (more conservative, better for small datasets)
- **Compare:** Report metrics for both splits; discuss which provides better generalization
- **Always:** Stratified splitting to maintain class distribution
- **References:**
- Hastie T, Tibshirani R, Friedman J "The Elements of Statistical Learning" (2009)
- Vabalas A et al. "Machine learning algorithm validation with a limited sample size" PLoS ONE (2019)

3.7 Scaling Strategy

Decision: Multiple scalers, compare

- **StandardScaler:** Primary (zero mean, unit variance) — for SVM, Logistic Regression, KNN, MLP
- **RobustScaler:** For datasets with outliers (uses IQR)
- **MinMaxScaler:** For tree-based models (optional, trees are scale-invariant)
- **Fit on training set ONLY**, transform both train and test
- **Reference:** Zheng A & Casari A "Feature Engineering for Machine Learning" O'Reilly (2018)

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4. FEATURE ENGINEERING (60+ Features)

> **CRITICAL:** Create ALL possible clinically meaningful features. Phase 1 creates them; Phase 2 evaluates and refines.

4.1 Ratio Features (Hormonal)

Feature Name	Formula	Clinical Rationale
FSH_LH_Ratio	FSH / LH	Key indicator of gonadal function; elevated ratio suggests primary testicular failure
Testosterone_FSH_Ratio	Testosterone / FSH	Lower ratio = worse spermatogenesis prognosis
Testosterone_LH_Ratio	Testosterone / LH	Leydig cell function indicator
LH_FSH_Ratio	LH / FSH	Inverse gonadal indicator
E2_Testosterone_Ratio	E2 / Testosterone	Estrogen dominance indicator
Prolactin_Testosterone_Ratio	Prolactin / Testosterone	Hyperprolactinemia impact
FSH_Testosterone_Ratio	FSH / Testosterone	Combined gonadal stress

4.2 Ratio Features (Testicular)

Feature Name	Formula	Clinical Rationale
RT_LT_Volume_Ratio	Testicular_volume_RT / Testicular_volume_LT	Testicular asymmetry indicator
RT_LT_Orchidometer_Ratio	RT_Size_Orchidometer / LT_Size_Orchidometer	Clinical asymmetry
RT_LT_Sakamoto_Ratio	Sakamoto_RT / Sakamoto_LT	Ultrasound-based asymmetry
Total_Testicular_Volume	Testicular_volume_RT + Testicular_volume_LT	Total gonadal volume
Mean_Testicular_Volume	(volume_RT + volume_LT) / 2	Average testicular size
Total_Sakamoto	Sakamoto_RT + Sakamoto_LT	Total Sakamoto score
Mean_Sakamoto	(Sakamoto_RT + Sakamoto_LT) / 2	Average density
Volume_Difference		volume_RT - volume_LT

4.3 Anthropometric Ratios

Feature Name	Formula	Clinical Rationale
BMI_Age_Interaction	BMI × Age	Metabolic-age combined effect
Weight_Height_Ratio	Weight / Height	Alternative to BMI

4.4 Clinical Binary Indicators

Feature Name	Condition	Clinical Rationale
Low_Testosterone	Testosterone < 300 ng/dL	Hypogonadism indicator
High_FSH	FSH > 12 IU/L	Primary testicular failure
Very_High_FSH	FSH > 20 IU/L	Severe testicular failure
Normal_Karyotype	Karyotype == “46,XY”	Genetic normality flag
Klinefelter	Karyotype contains “47,XXY”	Klinefelter syndrome
AZF_Deletion_Present	Y_microdeletion != “Normal” / “None”	AZF deletion presence
AZFc_Only	Y_microdeletion == “AZFc”	Isolated AZFc (better prognosis)
Elevated_Prolactin	Prolactin > 25 ng/mL	Hyperprolactinemia
Low_E2	E2 < 10 pg/mL	Estrogen deficiency
Has_Varicocele	Varicocele != null/none	Varicocele presence
Has_Surgery_History	Surgery_trauma != null/none	Prior surgical intervention
Has_Orchiopexy	Orchiopexy != null/none	History of undescended testis
Small_Testes_RT	Testicular_volume_RT < 4 mL	Right testicular atrophy
Small_Testes_LT	Testicular_volume_LT < 4 mL	Left testicular atrophy
Bilateral_Small_Testes	Both RT and LT < 4 mL	Severe bilateral atrophy

4.5 Age-Based Features

Feature Name	Definition	Reference
Age_Group	<35=Young, 35-39=Middle, 40-44=APA, ≥45=Elderly	AUA/ASRM guidelines
Age_Binary_40	Age < 40 vs ≥ 40	Most common binary cutoff in NOA literature
Age_Squared	Age^2	Non-linear age effect
Partner_Age_Group	Similar categorization	Partner age impact
Age_Difference	Age - Partner_Age	Couple age gap

4.6 Interaction Features

Feature Name	Formula	Clinical Rationale
Age_x_FSH	Age × FSH	Age-modulated hormonal effect
Age_x_Testosterone	Age × Testosterone	Age-hormone interaction
BMI_x_Testosterone	BMI × Testosterone	Metabolic-hormonal axis
FSH_x_TesticularVolume	FSH × Mean_Volume	Hormonal-structural interaction
Age_x_Volume	Age × Mean_Volume	Age-structural interaction
Karyotype_x_FSH	Normal_Karyotype × FSH	Genetic-hormonal interaction

4.7 Polynomial Features

Feature Name	Formula	Rationale
Age_Squared	Age^2	Capture non-linear age effects
FSH_Squared	FSH^2	Non-linear hormonal effect
BMI_Squared	BMI^2	Non-linear metabolic effect
Testosterone_Squared	$Testosterone^2$	Non-linear hormonal effect
Volume_Squared	$Mean\_Volume^2$	Non-linear size effect

4.8 Domain-Specific Composite Scores

Feature Name	Formula	Clinical Rationale
Hormonal_Score	Weighted combination of FSH, LH, Testosterone, E2, Prolactin	Overall endocrine status
Testicular_Score	Weighted combination of volume, Sakamoto, orchidometer	Overall testicular health
Genetic_Risk_Score	Combine Karyotype + AZF status	Overall genetic risk
Surgical_History_Score	Count of previous surgeries	Surgical burden
Comorbidity_Score	Sum of Diabetes + Hypertension + other comorbidities	Overall health status
Favorable_Prognostic_Score	Composite of positive predictors	Overall prognosis indicator

4.9 Statistical Transformations

Transformation	Applied To	Rationale
Log transformation	Right-skewed numerical (FSH, LH, Testosterone, Prolactin)	Normalize distributions
Square root	Count variables	Variance stabilization
Box-Cox	All continuous numerical	Optimal normalization

4.10 Missingness Features

Feature Name	Definition	Rationale
{col}_missing	1 if original value was missing, 0 otherwise	Missingness may itself be informative
Missing_Count	Total missing values per patient	Overall data completeness
Hormonal_Missing_Count	Missing count in hormonal features	Hormonal data availability
Testicular_Missing_Count	Missing count in testicular features	Structural data availability

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5. MODEL SELECTION (20+ Models)

> **All models must have scientific justification for inclusion.**

5.1 Classical Machine Learning Models (10 models)

#	Model	Library	Why Include	Key Considerations
1	**Logistic Regression**	sklearn	Interpretable baseline; gold standard for clinical prediction	Regularization: L1, L2, ElasticNet
2	**Decision Tree**	sklearn	Full interpretability; generates clinical rules	Prone to overfitting; use max_depth
3	**Random Forest**	sklearn	Robust ensemble; built-in feature importance	n_estimators, max_depth tuning
4	**XGBoost**	xgboost	State-of-art for tabular data; handles imbalance	scale_pos_weight, regularization
5	**LightGBM**	lightgbm	Faster than XGBoost; memory efficient	is_unbalance parameter
6	**CatBoost**	catboost	Native categorical handling; ordered boosting	auto_class_weights
7	**SVM**	sklearn	Excellent for moderate-sized data; kernel trick	Kernel: RBF, linear; C, gamma
8	**KNN**	sklearn	Non-parametric; intuitive	n_neighbors, distance metric
9	**Naive Bayes**	sklearn	Fast baseline; works with small data	GaussianNB, BernoulliNB
10	**Gradient Boosting**	sklearn	Standard sklearn boosting	n_estimators, learning_rate

5.2 Additional Tree-Based Models (2 models)

#	Model	Library	Why Include
11	**Extra Trees**	sklearn	More randomization than RF; may reduce overfitting
12	**AdaBoost**	sklearn	Adaptive boosting; good for weak learners

5.3 Deep Learning Models (3 models)

#	Model	Library	Why Include
13	**MLP (Multi-Layer Perceptron)**	sklearn/PyTorch	Universal approximator; neural baseline
14	**TabNet**	pytorch-tabnet	Attention-based; interpretable deep learning for tabular
15	**Wide & Deep**	Custom/TF	Combines memorization and generalization (optional if time permits)

5.4 Ensemble Methods (4 models)

#	Model	Library	Why Include
16	**Soft Voting Ensemble**	sklearn	Combine top-5 models by probability averaging
17	**Hard Voting Ensemble**	sklearn	Majority vote from top-5 models
18	**Stacking Ensemble**	sklearn	Meta-learner on base model predictions
19	**Blending Ensemble**	Custom	Holdout-based stacking alternative

5.5 Specialized Models (2+ models)

#	Model	Library	Why Include
20	**Elastic Net**	sklearn	Combines L1+L2; good for correlated features
21	**Ridge Classifier**	sklearn	L2-regularized linear model
22	**Balanced Random Forest**	imblearn	Imbalance-aware RF variant
23	**EasyEnsemble**	imblearn	Ensemble of under-sampled subsets

Total: 20-23 models (depending on feasibility)

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6. HYPERPARAMETER TUNING

6.1 Primary Method: Optuna (Bayesian Optimization)

Why Optuna?

- **Bayesian optimization** is more efficient than Grid Search (exhaustive but expensive) and Random Search (better than grid but still random)
- **Optuna advantages:**
- Tree-structured Parzen Estimator (TPE) sampler → intelligent search
- Pruning of unpromising trials → faster convergence
- Built-in visualization of optimization history
- Handles conditional hyperparameters elegantly
- **References:**
- Akiba T et al. “Optuna: A Next-generation Hyperparameter Optimization Framework” KDD (2019)
- Bergstra J et al. “Algorithms for Hyper-Parameter Optimization” NIPS (2011)
- Snoek J et al. “Practical Bayesian Optimization of Machine Learning Algorithms” NIPS (2012)

Configuration

- **Trials per model:** 100-200 (depending on model complexity)
- **Sampler:** TPE (Tree-structured Parzen Estimator)
- **Pruner:** MedianPruner (prune after 5 warmup steps)
- **Objective:** Maximize AUC-ROC (primary), with F1 as secondary
- **CV within tuning:** 5-fold stratified cross-validation

Comparison Requirement

- **Also run:** Grid Search and Random Search on 2-3 representative models (e.g., XGBoost, Random Forest)
- **Document:** Show that Optuna converges faster and finds better parameters
- **This comparison itself is a contribution** if properly documented

6.2 Hyperparameter Spaces (Per Model)

Logistic Regression

```
C: [0.001, 0.01, 0.1, 1, 10, 100]
penalty: ['l1', 'l2', 'elasticnet']
solver: ['lbfgs', 'saga', 'liblinear']
max_iter: [1000, 5000, 10000]
```

Random Forest

```
n_estimators: [100, 200, 300, 500, 1000]
max_depth: [3, 5, 7, 10, 15, 20, None]
min_samples_split: [2, 5, 10, 20]
min_samples_leaf: [1, 2, 4, 8]
max_features: ['sqrt', 'log2', 0.3, 0.5, 0.7]
class_weight: ['balanced', 'balanced_subsample']
```

XGBoost

```
n_estimators: [100, 200, 300, 500, 1000]
max_depth: [3, 5, 7, 9, 11]
learning_rate: [0.01, 0.05, 0.1, 0.2, 0.3]
min_child_weight: [1, 3, 5, 7]
subsample: [0.6, 0.7, 0.8, 0.9, 1.0]
colsample_bytree: [0.6, 0.7, 0.8, 0.9, 1.0]
gamma: [0, 0.1, 0.2, 0.5, 1]
reg_alpha: [0, 0.001, 0.01, 0.1, 1]
reg_lambda: [0, 0.001, 0.01, 0.1, 1]
scale_pos_weight: [auto-calculated from class ratio]
```

SVM

```
C: [0.01, 0.1, 1, 10, 100]
kernel: ['rbf', 'linear', 'poly', 'sigmoid']
gamma: ['scale', 'auto', 0.001, 0.01, 0.1, 1]
class_weight: ['balanced']
probability: [True]  # Required for AUC calculation
```

LightGBM

```
n_estimators: [100, 200, 300, 500, 1000]
max_depth: [-1, 3, 5, 7, 9]
learning_rate: [0.01, 0.05, 0.1, 0.2]
num_leaves: [15, 31, 63, 127]
min_child_samples: [5, 10, 20, 50]
subsample: [0.6, 0.7, 0.8, 0.9, 1.0]
colsample_bytree: [0.6, 0.7, 0.8, 0.9, 1.0]
is_unbalance: [True]
```

CatBoost

```
iterations: [100, 200, 500, 1000]
depth: [4, 6, 8, 10]
learning_rate: [0.01, 0.05, 0.1, 0.2]
l2_leaf_reg: [1, 3, 5, 7, 9]
auto_class_weights: ['Balanced', 'SqrtBalanced']
```

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7. VALIDATION STRATEGY

> **CRITICAL:** This is what reviewers scrutinize most. Must be comprehensive.

7.1 Primary Validation Methods

#	Method	Configuration	Purpose
1	**10-Fold Stratified CV**	10 folds, stratified, multiple seeds	Standard validation gold standard
2	**5-Fold Stratified CV**	5 folds, stratified	Comparison with 10-fold
3	**Leave-One-Out CV (LOOCV)**	N iterations (N=2450)	Maximum use of data; unbiased but high variance
4	**Nested CV**	Outer: 5-fold, Inner: 3-fold	Unbiased performance estimation WITH hyperparameter tuning
5	**Repeated K-Fold**	10-fold × 10 repeats = 100 total folds	Stability across random splits
6	**Bootstrap Validation**	1000 iterations with replacement	95% confidence intervals for all metrics
7	**80/20 Stratified Split**	Primary holdout evaluation	Standard split
8	**70/30 Stratified Split**	Secondary holdout evaluation	Conservative split comparison

7.2 Robustness Testing

#	Test	Configuration	Purpose
9	**Noise Injection**	Add 5%, 10%, 15% Gaussian noise	Performance degradation analysis
10	**Feature Perturbation**	Randomly shuffle one feature at a time	Feature sensitivity analysis
11	**Stability Analysis**	Train on 8 different random seeds	Seed sensitivity
12	**Learning Curves**	Train sizes: 10%, 20%, 30%, …, 100%	Bias-variance tradeoff

7.3 Overfitting Detection

#	Check	Threshold	Action
13	**Train-Test AUC Gap**	Gap > 5% → suspect	Apply regularization, reduce complexity
14	**Cross-Validation Variance**	SD > 0.05 → unstable	Simplify model, increase regularization
15	**Perfect Train AUC (1.0)**	Always suspicious	Create adjusted/regularized model variant

7.4 Statistical Significance Testing

#	Test	Purpose
16	**DeLong Test**	Compare AUCs between models (DeLong et al., 1988)
17	**McNemar’s Test**	Compare classification accuracy between models
18	**Paired t-test on CV folds**	Compare metrics across CV folds
19	**Friedman Test**	Compare multiple classifiers simultaneously

7.5 Temporal Validation (if time column available)

- **Decision:** If surgery dates are available in the dataset, split by time (train on earlier cases, test on later)
- **If no time column:** Note as limitation; recommend for future work

7.6 Metrics to Report for ALL Models

Metric	Formula/Description	Why
AUC-ROC	Area Under ROC Curve	Primary metric — discrimination ability
Accuracy	(TP+TN)/(TP+TN+FP+FN)	Overall correctness
Sensitivity (Recall)	TP/(TP+FN)	Clinically critical — don't miss positive cases
Specificity	TN/(TN+FP)	Avoid false positives
Precision (PPV)	TP/(TP+FP)	Positive predictive value
NPV	TN/(TN+FN)	Negative predictive value
F1 Score	2×(Precision×Recall)/(Precision+Recall)	Balance of precision and recall
MCC	Matthews Correlation Coefficient	Best single metric for imbalanced data
Brier Score	Mean squared difference between predicted prob and actual	Calibration quality
Cohen's Kappa	Agreement beyond chance	Inter-rater agreement
Log Loss	Negative log-likelihood	Probability quality
Balanced Accuracy	(Sensitivity + Specificity) / 2	Imbalance-adjusted accuracy

8. EXPLAINABLE AI (XAI) METHODS

8.1 Feature-Level Explanations

#	Method	Scope	Models	Details
1	**Native Feature Importance**	Global	All tree-based + Logistic Regression	`feature_importances_` / `|coef_|`, normalized
2	**Permutation Importance**	Global	All models	30 repeats, `roc_auc` scorer
3	**SHAP (Global)**	Global	XGBoost, RF, LightGBM, CatBoost, SVM, LR	TreeExplainer for trees, KernelExplainer for others; beeswarm + bar plots
4	**SHAP (Local)**	Local	Top-5 models	Waterfall plots for 3 archetypes: High-Positive, High-Negative, Borderline
5	**SHAP Dependence**	Global	Top-5 models	Top-4 features per model with interaction coloring
6	**SHAP Interaction**	Global	XGBoost	SHAP interaction values matrix
7	**PDP (Partial Dependence)**	Global	XGBoost, RF	Top-4 features, 2×2 grid
8	**ICE (Individual Conditional)**	Global+Local	XGBoost, RF	ICE + PDP overlay, top-4 features
9	**LIME**	Local	XGBoost, SVM (best model)	5 archetype samples: High-Pos, High-Neg, Borderline, True-Pos, False-Neg
10	**ELI5**	Global	All models	Quick importance + weights overview
11		Global	Top-3 models	Alternative to PDP that

#	Method	Scope	Models	Details
	Accumulated Local Effects (ALE)			handles correl-ated features

8.2 Model Performance Visualization

#	Visualization	Details
12	**ROC Curves (Standard)**	All models overlaid on single plot with AUC in legend
13	**ROC Curves (with 95% CI)**	Bootstrap 1000 iterations; shaded CI bands; per-model subplot
14	**Precision-Recall Curves**	All models with Average Pre-cision; no-skill baseline
15	**Calibration Curves**	Reliability diagrams (10 bins); perfect calibration reference
16	**Confusion Matrices**	All models at threshold=0.5 AND at optimal threshold (from Youden's J)
17	**Decision Curve Analysis (DCA)**	Clinical utility across threshold probabilities
18	**Net Benefit Curves**	Compared to "treat all" and "treat none" strategies

8.3 Clinical Synthesis

#	Deliverable	Details
19	**Consensus Feature Ranking**	Aggregate top features across all models + all XAI methods
20	**Clinical Interpretation Report**	In English + Farsi; what features matter, clinical implications
21	**Final Research Findings**	Best model, top predictors, clinical recommendations
22	**Literature Comparison**	Compare results with published NOA prediction studies

8.4 Sample Archetype Definitions for Local XAI

Archetype	Selection Criterion
High-Positive	Highest predicted probability of success
High-Negative	Lowest predicted probability (near-zero)
Borderline	Predicted probability closest to 0.5
True-Positive	Actual=1 AND predicted probability > 0.7
False-Negative	Actual=1 AND predicted probability < 0.3

9. OUTPUT REQUIREMENTS

9.1 Report Files

#	File	Location	Description
1	`Phase{N}_Report.md`	`3_Results/Phase{N}/`	Detailed technical report per phase
2	`clinical_interpretation.md`	`5_Reports/`	Clinical interpretation (English + Farsi)
3	`Final_Research_Findings.md`	`5_Reports/`	Final summary for publication
4	`Decision_Logger.md`	`5_Reports/`	ALL decisions with references — for thesis writing
5	`Literature_Comparison.md`	`5_Reports/`	Comparison with published NOA ML studies
6	`Executive_Summary.md`	`5_Reports/`	Non-technical summary
7	`Phase{N}_Summary_FA.md`	`5_Reports/`	Farsi summary per phase
8	`outlier_report.csv`	`3_Results/Phase1/`	Detected outliers for clinical review
9	`Overall_Comprehensive_Report.md`	`5_Reports/`	Single consolidated report across all phases

9.2 Decision Logger Requirements

The `Decision_Logger.md` must contain for EVERY decision:
1. **Decision ID** (sequential)
2. **Phase** (which phase)
3. **Category** (e.g., Missing Data, Feature Selection, Model Selection)
4. **Decision** (what was decided)
5. **Rationale** (why)
6. **Reference** (APA format citation)
7. **Alternative Considered** (what else was considered)

8. **Novelty Flag** (if our approach outperformed the reference method)
9. **Outcome** (what happened after implementing)

9.3 Data Files

#	File	Location	Description
1	`original_dataset.csv`	`1_Data/Raw/`	Original converted from .xlsb
2	`cleaned_dataset.csv`	`1_Data/Cleaned/`	After cleaning (Phase 1)
3	`encoded_dataset.csv`	`1_Data/Processed/`	After encoding + scaling (Phase 1)
4	`feature_engineered_dataset.csv`	`1_Data/Processed/`	After feature engineering
5	`descriptive_stats.csv`	`3_Results/Phase2/`	Descriptive statistics
6	`normality_tests.csv`	`3_Results/Phase2/`	Normality test results
7	`group_comparison.csv`	`3_Results/Phase2/`	Group comparison results
8	`correlation_matrix.csv`	`3_Results/Phase2/`	Correlation matrices
9	`model_comparison.csv`	`3_Results/Phase3/`	All model metrics comparison
10	`hyperparameter_results.csv`	`3_Results/Phase3/`	Optuna tuning results
11	`optimal_thresholds.csv`	`3_Results/Phase3/`	Per-model optimal thresholds
12	`kfold_results.csv`	`3_Results/Phase4/`	K-Fold CV results
13	`loocv_results.csv`	`3_Results/Phase4/`	LOOCV results
14	`bootstrap_results.csv`	`3_Results/Phase4/`	Bootstrap validation results
15	`delong_test_results.csv`	`3_Results/Phase4/`	DeLong test results

9.4 Model Files

Location	Format	Description
`6_Models/Saved/`	`.joblib`	All trained models (base + adjusted)
`6_Models/Saved/`	`.json`	Hyperparameter configurations
`6_Models/Saved/`	`.pkl`	Scalers, encoders

9.5 Figure Specifications

- **Format:** BOTH PNG (for reports) AND TIFF (for journal submission)
- **DPI:** 300 minimum (publication quality)
- **Location:** `4_Figures/Phase{N}/`
- **Naming:** `{figure_type}_{model_name}.{png|tiff}`
- **Color scheme:** Consistent across all figures (use seaborn/plotly professional palettes)
- **Font size:** Minimum 12pt for labels, 10pt for tick marks
- **Figure size:** Minimum 10×8 inches for single plots, 14×12 for grids

9.6 Literature Comparison Requirements

The `Literature_Comparison.md` must include:
1. **Search:** PubMed/Google Scholar for "NOA sperm retrieval prediction machine learning" (2015-2025)
2. **Table:** Compare our results with published studies:
- Study (Author, Year)
- Sample Size
- Features Used
- Models Used
- Best AUC
- Our AUC
- Key Differences
3. **Discussion:** Where we improve, where we're comparable, limitations
4. **Novelty Claims:** What's new in our approach

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10. DOUBLE-CHECK POINTS

> **After EACH phase, before proceeding, perform these checks:**

10.1 After Phase 1 (Data Cleaning)

#	Check	How
1	No data leakage	Verify target column is not used in feature engineering
2	Missing values handled correctly	Verify imputation was on train only
3	No duplicate rows	`df.duplicated().sum() == 0`
4	Data types correct	All numerical features are float/int, categoricals encoded
5	Outlier report generated	`outlier_report.csv` exists with all flagged outliers
6	Feature engineering produces valid values	No NaN, no Inf in engineered features
7	Target distribution preserved	Class ratio matches original after cleaning
8	Column exclusions documented	Verify correct columns were dropped
9	Shape is reasonable	n_samples matches expectation after row drops
10	Decision logger updated	All Phase 1 decisions logged with references

10.2 After Phase 2 (Statistical Analysis)

#	Check	How
1	Normality tests run for all numerical	Shapiro-Wilk, K-S results logged
2	Correct test selection	Parametric vs non-parametric based on normality
3	VIF calculated	Multicollinearity check; flag VIF > 10
4	Correlations computed	Pearson AND Spearman (based on normality)
5	Group comparisons significant features	List features with $p < 0.05$ between outcome groups
6	Power analysis adequate	Statistical power ≥ 0.80
7	Class imbalance documented	Ratio, visualization, strategy defined
8	Effect sizes reported	Cohen’s d / rank-biserial for all comparisons
9	Feature selection results	Which features survived each stage
10	Decision logger updated	All Phase 2 decisions logged with references

10.3 After Phase 3 (Model Training)

#	Check	How
1	All models trained	20+ models × 8 seeds = 160+ runs
2	SMOTE on train only	Verify SMOTE was NOT applied to test data
3	Scaling on train only	Verify scaler fit on train, transform on test
4	Optuna completed	100-200 trials per model
5	Model comparison table	All metrics for all models
6	No impossible metrics	No AUC = 1.0 on test set (suspicious)
7	Models saved	All `.joblib` files exist
8	Hyperparameters documented	All final parameters logged
9	Ensemble models created	Voting + Stacking at minimum
10	Decision logger updated	All Phase 3 decisions logged with references

10.4 After Phase 4 (Validation)

#	Check	How
1	All CV methods run	10-fold, 5-fold, LOOCV, Nested, Repeated, Bootstrap
2	Both splits tested	80/20 AND 70/30 results
3	Overfitting detected and addressed	Train-test gap < 5% for final models
4	Learning curves plotted	Bias-variance analyzed
5	Stability verified	Seed sensitivity < 2% variation
6	Robustness tested	Performance under noise documented
7	Statistical significance	DeLong test between top models
8	Adjusted models created	For any overfitting models
9	Bootstrap CIs	95% CI for all metrics
10	Decision logger updated	All Phase 4 decisions logged with references

10.5 After Phase 5 (XAI)

#	Check	How
1	SHAP for all required models	TreeExplainer + KernelExplainer as needed
2	LIME for multiple models	At least XGBoost + best model
3	PDP/ICE plots generated	Top-4 features per model
4	ROC with CI	Bootstrap 1000 iterations
5	Confusion matrices at optimal threshold	Not just 0.5
6	Clinical interpretation written	English + Farsi
7	Literature comparison complete	At least 10 published studies compared
8	Consensus feature ranking	Aggregated across all models & XAI methods
9	All figures publication-quality	300 DPI, PNG + TIFF
10	Decision logger complete	ALL decisions across all phases

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11. PHASE-BY-PHASE BREAKDOWN

11.1 PHASE 0: Project Setup & Initialization

Tasks

1. Create standardized folder structure
2. Install all Python dependencies
3. Create `requirements.txt`
4. Convert `.xlsb` to `.csv` and save to `1_Data/Raw/`
5. Initial dataset preview (shape, columns, dtypes)
6. Initialize Decision Logger

Inputs

- Raw dataset: `[path] Rad_Final ID.xlsb`
- This specification document

Outputs

```
NOA_ML_Project/
├── 1_Data/
│   ├── Raw/original_dataset.csv
│   ├── Cleaned/          (empty, populated in Phase 1)
│   └── Processed/        (empty, populated in Phase 1)
├── 2_Code/
├── 3_Results/
│   └── Phase0/Phase0_Setup_Report.md
├── 4_Figures/
├── 5_Reports/
│   └── Decision_Logger.md    (initialized)
├── 6_Models/
│   └── Saved/
└── requirements.txt
```

Folder Structure

```
NOA_ML_Project/
├── 1_Data/
│   ├── Raw/                    # Original unchanged data
│   ├── Cleaned/                # After Phase 1 cleaning
│   └── Processed/              # After encoding, scaling, feature engineering
├── 2_Code/
│   ├── phase0_setup.py
│   ├── phase1_cleaning.py
│   ├── phase2_statistics.py
│   ├── phase3_modeling.py
│   ├── phase4_validation.py
│   └── phase5_xai.py
├── 3_Results/
│   ├── Phase0/
│   ├── Phase1_DataCleaning/
│   ├── Phase2_Statistics/
│   ├── Phase3_Models/
│   ├── Phase4_Validation/
│   └── Phase5_XAI/
├── 4_Figures/
│   ├── Phase1/
│   ├── Phase2/
│   ├── Phase3/
│   ├── Phase4/
│   └── Phase5/
├── 5_Reports/
│   ├── Decision_Logger.md
│   ├── clinical_interpretation.md
│   ├── Final_Research_Findings.md
│   ├── Literature_Comparison.md
│   └── Overall_Comprehensive_Report.md
├── 6_Models/
│   └── Saved/
└── requirements.txt
```

11.2 PHASE 1: Data Understanding & Cleaning

Tasks (10 tasks)

Task 1: Read & Understand Dataset
- Load `.xlsb` file, identify all columns, dtypes, shape, memory
- Preview first/last rows, check for obvious issues
- Document: column count, row count, target variable

Task 2: Intelligent Data Type Detection
- Classify each column: Binary, Categorical (Nominal/Ordinal), Numerical (Continuous/Discrete), Text, DateTime
- Create `data_type_classification.csv`

Task 3: Missing Values Analysis & Imputation
- Calculate missing % per column
- Apply >50% threshold (with clinical exceptions) — Reference: EMA ICH E9(R1)
- Create missingness indicator variables
- Apply MissForest imputation — Reference: Stekhoven & Bühlmann (2012)
- Fallback: MICE if MissForest fails
- Document every imputation decision with reference

Task 4: Outlier Detection & Documentation
- IQR method for all numerical features — Reference: Aguinis et al. (2013)
- Z-score (|Z| > 3) for confirmation
- Generate `outlier_report.csv` — DO NOT auto-remove
- Winsorize (cap at 1st/99th percentile) unless clinical review says otherwise
- Flag for medical team review

Task 5: Duplicate & Inconsistency Check
- Check exact duplicates
- Check near-duplicates (same patient, different rows)
- Verify logical consistency (e.g., BMI matches Height/Weight)

Task 6: Data Type Corrections
- Convert `object` columns that should be numeric (e.g., Testosterone stored as string)
- Parse compound values (e.g., ranges like “10-15” → midpoint)
- Handle encoding issues (Farsi characters, special symbols)

Task 7: Feature Engineering (see Section 4)
- Create ALL 60+ features from Section 4
- All ratio features, binary indicators, interactions, polynomial, composite scores
- Missingness indicator features
- Statistical transformations (log, sqrt, Box-Cox)

Task 8: Encoding Categorical Variables
- Label Encoding for ordinal variables
- One-Hot Encoding for nominal variables with ≤10 categories
- Target Encoding for high-cardinality categorical (Pathology-RT with 772 values)
- Reference: Potdar et al. “A Comparative Study of Categorical Variable Encoding Techniques” (2017)

Task 9: Scaling Numerical Features
- StandardScaler as primary

- RobustScaler as alternative for features with outliers
- **FIT ON TRAINING SET ONLY**

Task 10: Save & Report
- Save `cleaned_dataset.csv`, `encoded_dataset.csv`, `feature_engineered_dataset.csv`
- Generate `Phase1_Report.md` with all decisions documented
- Update Decision Logger

Inputs

- `1_Data/Raw/original_dataset.csv`

Outputs

- `1_Data/Cleaned/cleaned_dataset.csv`
- `1_Data/Processed/encoded_dataset.csv`
- `1_Data/Processed/feature_engineered_dataset.csv`
- `3_Results/Phase1_DataCleaning/Phase1_Report.md`
- `3_Results/Phase1_DataCleaning/outlier_report.csv`
- `3_Results/Phase1_DataCleaning/data_type_classification.csv`
- `3_Results/Phase1_DataCleaning/missing_values_analysis.csv`
- `4_Figures/Phase1/missing_values_analysis.{png,tiff}`
- `4_Figures/Phase1/outliers_boxplot.{png,tiff}`
- `4_Figures/Phase1/data_types_summary.{png,tiff}`
- `4_Figures/Phase1/correlation_initial.{png,tiff}`

11.3 PHASE 2: Advanced Statistical Analysis

Tasks (8 tasks)

Task 1: Comprehensive Descriptive Statistics
- Mean, SD, Median, IQR, 95% CI — overall AND by target group (Success vs. Failure)
- For categorical: frequencies, proportions, chi-square tests
- Output: `descriptive_stats.csv`

Task 2: Distribution / Normality Analysis
- Shapiro-Wilk, Kolmogorov-Smirnov, D'Agostino tests
- Q-Q plots, histograms per feature
- Skewness and kurtosis for all numerical features
- This determines parametric vs non-parametric test selection
- Output: `normality_tests.csv`

Task 3: Correlation Analysis
- Pearson (for normally distributed numerical features)
- Spearman (for non-normal / ordinal features)
- Point-biserial (for binary-continuous pairs)
- Heatmaps for both Pearson and Spearman
- Output: `correlation_matrix.csv`

Task 4: Group Comparison Tests
- For numerical features: Independent t-test (if normal) OR Mann-Whitney U (if non-normal)

- For categorical features: Chi-square test OR Fisher's exact test
- Report: test statistic, p-value, effect size (Cohen's d or rank-biserial correlation)
- Bonferroni correction for multiple comparisons
- Output: `group_comparison.csv`

Task 5: ANOVA / Kruskal-Wallis
- For multi-group comparisons (e.g., Age Groups vs. outcome)
- Post-hoc: Tukey HSD or Dunn's test
- Report effect sizes (η^2)

Task 6: Statistical Power Analysis
- Calculate achieved power for each significant feature
- Report minimum detectable effect size
- Ensure power ≥ 0.80 for primary comparisons
- Reference: Cohen J "Statistical Power Analysis for the Behavioral Sciences" (1988)

Task 7: Class Imbalance Analysis
- Visualize class distribution
- Calculate imbalance ratio
- Document SMOTE + Class Weights strategy (see Section 3.2)
- Test SMOTE variants (ADASYN, Borderline-SMOTE, SMOTETomek) and document

Task 8: Multicollinearity Check (VIF)
- Calculate Variance Inflation Factor for all features
- Flag features with VIF > 10 (severe multicollinearity)
- Flag features with VIF > 5 (moderate multicollinearity)
- Decision: Remove or combine correlated features
- Reference: James G et al. "ISLR" (2013)

Inputs

- `1_Data/Cleaned/cleaned_dataset.csv`
- `1_Data/Processed/feature_engineered_dataset.csv`
- `Phase1_Report.md`

Outputs

- `3_Results/Phase2_Statistics/Phase2_Report.md`
- `3_Results/Phase2_Statistics/descriptive_stats.csv`
- `3_Results/Phase2_Statistics/normality_tests.csv`
- `3_Results/Phase2_Statistics/group_comparison.csv`
- `3_Results/Phase2_Statistics/correlation_matrix.csv`
- `3_Results/Phase2_Statistics/vif_analysis.csv`
- `3_Results/Phase2_Statistics/power_analysis.csv`
- `4_Figures/Phase2/distribution_analysis.{png,tiff}`
- `4_Figures/Phase2/pearson_heatmap.{png,tiff}`
- `4_Figures/Phase2/spearman_heatmap.{png,tiff}`
- `4_Figures/Phase2/class_distribution.{png,tiff}`
- `4_Figures/Phase2/qq_plots.{png,tiff}`
- `4_Figures/Phase2/group_comparison_boxplots.{png,tiff}`

11.4 PHASE 3: Intelligent Model Selection & Training

Tasks (8 tasks)

Task 1: Dataset Analysis for Model Selection
- Analyze: sample size, feature count, sample-to-feature ratio, class balance
- Auto-recommend models based on dataset characteristics
- Document reasoning

Task 2: Data Preparation
- Train/test split: 80/20 stratified (primary) AND 70/30 stratified (secondary)
- Apply StandardScaler (fit on train only)
- Apply SMOTE on training set only
- Save splits for reproducibility

Task 3: Class Imbalance Handling
- Apply SMOTE + Class Weights (Section 3.2)
- Also test: ADASYN, SMOTETomek, Borderline-SMOTE
- Document which works best with each model

Task 4: Define & Justify All Models
- All 20+ models from Section 5
- For each: scientific justification, strengths, weaknesses
- Reference for each model selection decision

Task 5: Hyperparameter Tuning (Optuna)
- 100-200 Optuna trials per model (Section 6)
- Also run Grid/Random on 2-3 models for comparison
- Document Optuna vs Grid vs Random Search results
- Save Optuna study objects for visualization

Task 6: Train & Evaluate All Models
- Train all models with optimized hyperparameters
- Evaluate on test set: all metrics from Section 7.6
- Generate model comparison table
- Report with 8 random seeds (mean ± SD)

Task 7: Ensemble Methods
- Build Voting (Soft + Hard), Stacking, Blending ensembles
- Use top-5 base models as ensemble components
- Evaluate ensembles with same metrics

Task 8: Deep Learning (MLP, TabNet)
- Train MLP and TabNet
- Compare with classical models
- Document if deep learning provides improvement

Inputs

- `1_Data/Processed/encoded_dataset.csv` (or feature_engineered version)
- `3_Results/Phase2_Statistics/Phase2_Report.md`

Outputs

- `3_Results/Phase3_Models/Phase3_Report.md`

- `3_Results/Phase3_Models/model_comparison.csv`
- `3_Results/Phase3_Models/hyperparameter_results.csv`
- `3_Results/Phase3_Models/optimal_thresholds.csv`
- `3_Results/Phase3_Models/optuna_vs_grid_random.csv`
- `6_Models/Saved/*.joblib` (all trained models)
- `6_Models/Saved/hyperparameters.json`
- `4_Figures/Phase3/model_comparison_bar.{png,tiff}`
- `4_Figures/Phase3/optuna_optimization_history.{png,tiff}`

11.5 PHASE 4: Cross-Validation & Robustness Testing

Tasks (10 tasks)

Task 1: K-Fold Cross-Validation
- 5-Fold AND 10-Fold Stratified CV
- Report: mean ± SD for all metrics

Task 2: Stratified K-Fold
- Verify stratification maintains class distribution

Task 3: Leave-One-Out Cross-Validation (LOOCV)
- N=2450 iterations
- Report: accuracy, AUC (from collected predictions)

Task 4: Repeated K-Fold
- 10-Fold × 10 repeats = 100 total folds
- Provides most stable estimate

Task 5: Bootstrap Validation
- 1000 bootstrap iterations with replacement
- Report: mean, SD, 95% CI for all metrics

Task 6: Nested Cross-Validation
- Outer: 5-fold, Inner: 3-fold
- Unbiased estimate with hyperparameter tuning
- Compare with non-nested to quantify optimistic bias

Task 7: Overfitting Detection
- Train vs. test metric comparison
- Flag models with gap > 5%
- Create regularized/adjusted variants for overfitting models
- Save both original and adjusted models

Task 8: Learning Curves
- Train sizes: [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0]
- Plot training score vs validation score
- Diagnose: high bias vs high variance

Task 9: Stability Analysis
- Run all models with 8 different random seeds

- Report variation (SD) per model
- Flag models with SD > 2%

Task 10: Robustness Testing
- Inject 5%, 10%, 15% Gaussian noise
- Measure performance degradation
- Feature perturbation analysis
- Rank models by robustness

Inputs

- `1_Data/Processed/encoded_dataset.csv`
- `6_Models/Saved/*.joblib`
- `3_Results/Phase3_Models/Phase3_Report.md`

Outputs

- `3_Results/Phase4_Validation/Phase4_Report.md`
- `3_Results/Phase4_Validation/kfold_cv_results.csv`
- `3_Results/Phase4_Validation/loocv_results.csv`
- `3_Results/Phase4_Validation/bootstrap_results.csv`
- `3_Results/Phase4_Validation/nested_cv_results.csv`
- `3_Results/Phase4_Validation/overfitting_analysis.csv`
- `3_Results/Phase4_Validation/stability_analysis.csv`
- `3_Results/Phase4_Validation/robustness_results.csv`
- `3_Results/Phase4_Validation/learning_curves_data.csv`
- `3_Results/Phase4_Validation/delong_test_results.csv`
- `6_Models/Saved/*_adjusted.joblib` (regularized variants)
- `4_Figures/Phase4/learning_curves.{png,tiff}`
- `4_Figures/Phase4/bootstrap_distributions.{png,tiff}`
- `4_Figures/Phase4/stability_comparison.{png,tiff}`
- `4_Figures/Phase4/robustness_heatmap.{png,tiff}`
- `4_Figures/Phase4/overfitting_analysis.{png,tiff}`
- `4_Figures/Phase4/nested_cv_comparison.{png,tiff}`

11.6 PHASE 5: Explainable AI (XAI) & Clinical Interpretation

Tasks (10 tasks)

Task 1: Feature Importance (All Models)
- Native importance: `feature_importances_` for tree-based, `|coef_|` for linear
- Normalize to 0-100% scale
- Compare top-10 features across models
- Output: feature_importance_comparison figure

Task 2: Permutation Importance
- `sklearn.inspection.permutation_importance` (30 repeats, `roc_auc`)

- For ALL models
- Report: mean ± std per feature per model

Task 3: SHAP Analysis (Global & Local)
- **Global:** Summary beeswarm + bar plots per model
- **Local:** Waterfall plots for 3 archetypes (High-Pos, High-Neg, Borderline)
- **Dependence:** Top-4 features per model
- TreeExplainer: XGBoost, RF, LightGBM, CatBoost, GradientBoosting, ExtraTrees
- KernelExplainer: SVM, LogisticRegression
- **Include SVM** (best model from Phase 4 with synthetic data — may differ with real data)

Task 4: PDP & ICE Plots
- Partial Dependence: top-4 features, 2×2 grid per model
- ICE overlay: individual + average lines
- For: XGBoost, RandomForest, and best model

Task 5: LIME Explanations
- 5 archetype samples (High-Pos, High-Neg, Borderline, True-Pos, False-Neg)
- For: XGBoost + best model (not just XGBoost)
- `num_features=10`

Task 6: ROC Curves
- Standard: all models overlaid, AUC in legend
- With CI: bootstrap 1000 iterations, 95% CI shaded bands
- Individual per-model ROC subplot

Task 7: Precision-Recall Curves
- All models on single plot
- Average Precision (AP) in legend
- No-skill baseline at class prevalence

Task 8: Calibration Curves
- Reliability diagrams (10 bins)
- Perfect calibration reference line
- For all models

Task 9: Confusion Matrices
- At threshold = 0.5 AND at optimal threshold (Youden's J)
- Report: Sensitivity, Specificity, PPV, NPV per model
- Use clinical variable names (not Feature_N)

Task 10: Clinical Interpretation & Research Findings
- Consensus feature ranking across all models + all XAI methods
- Clinical implications (what do top features mean for patient counseling?)
- Best model recommendation with justification
- Limitations section
- Future work recommendations (including external validation, temporal validation, CDSS)
- Literature comparison with published NOA prediction studies
- **Write in both English and Farsi**

Inputs

- `1_Data/Processed/encoded_dataset.csv`

- `6_Models/Saved/*.joblib`
- `3_Results/Phase4_Validation/Phase4_Report.md`
- `feature_mapping.json` (Feature_N → clinical variable mapping)

Outputs

- `3_Results/Phase5_XAI/Phase5_Report.md`
- `5_Reports/clinical_interpretation.md` (English + Farsi)
- `5_Reports/Final_Research_Findings.md`
- `5_Reports/Literature_Comparison.md`
- `5_Reports/Decision_Logger.md` (complete)
- `5_Reports/Overall_Comprehensive_Report.md`
- 16+ figure files (PNG + TIFF) in `4_Figures/Phase5/`:
- `feature_importance_comparison`
- `shap_summary_{model}` (per model)
- `shap_bar_{model}` (per model)
- `shap_waterfall_{model}_{case}` (per model × 3 cases)
- `shap_dependence_{model}` (per model)
- `pdp_{model}` (per model)
- `ice_{model}` (per model)
- `lime_{model}_{case}` (per model × 5 cases)
- `roc_curves_comparison`
- `roc_curves_with_ci`
- `precision_recall_curves`
- `calibration_curves`
- `confusion_matrices`
- `confusion_matrices_optimal_threshold`
- `consensus_feature_ranking`

---

12. QUALITY REQUIREMENTS

12.1 Publication-Ready Standards

- All figures at ≥300 DPI in both PNG and TIFF
- All tables formatted for journal submission
- Statistical reporting follows APA/STARD/TRIPOD guidelines
- Confidence intervals reported for ALL primary metrics
- Effect sizes alongside p-values
- Sample size justification documented
- TRIPOD checklist compliance

12.2 Evidence-Based

- Every methodological decision cites a reference

- Prefer references from 2015-2025 (recent)
- Use landmark/seminal papers where appropriate (even if older)
- Decision Logger captures everything for thesis writing

12.3 Novelty Testing

- For every established method used, also test an alternative/novel approach
- If novel approach outperforms: document as a contribution
- If established method is better: document that we validated it
- This creates potential for publication novelty claims

12.4 Complete Documentation

- Every phase has its own report
- Overall comprehensive report at the end
- Decision Logger with ALL decisions + references
- Code is commented and documented
- Reproducibility: any researcher can re-run with the same results

12.5 Clinical Validity

- Use clinical variable names (not Feature_N) in all patient-facing reports
- Feature mapping JSON must be used to translate
- Clinical interpretation reviewed for medical accuracy
- Outlier decisions flagged for clinical team review

---

13. TECHNICAL SPECIFICATIONS

13.1 Python Environment

Required Packages

```
# Core
pandas>=2.0
numpy>=1.24
scipy>=1.10
scikit-learn>=1.3

# Gradient Boosting
xgboost>=2.0
lightgbm>=4.0
catboost>=1.2

# Deep Learning
pytorch-tabnet>=4.0
torch>=2.0

# Imbalanced Learning
imbalanced-learn>=0.11

# Imputation
missingpy  # MissForest
sklearn  # IterativeImputer (MICE)

# XAI
shap>=0.43
lime>=0.2
eli5>=0.13

# Optimization
optuna>=3.0

# Visualization
matplotlib>=3.7
seaborn>=0.12
plotly>=5.15

# Statistical
statsmodels>=0.14
pingouin  # Effect sizes, power analysis

# Utilities
joblib
tqdm
openpyxl
pyxlsb  # For reading .xlsb files
```

13.2 Hardware Requirements

- **RAM:** ≥16 GB (for LOOCV with 2450 iterations)
- **Storage:** ≥10 GB (for models, figures, results)
- **CPU:** Multi-core recommended (for parallel CV)

13.3 Random State Convention

- **Primary seed:** 42
- **All seeds:** [42, 123, 456, 789, 1024, 2023, 2024, 3141]
- Always set `random_state` in all functions that accept it

13.4 Data Leakage Prevention Checklist

- ☐ SMOTE applied ONLY to training set
- ☐ Scaler fit ONLY on training set
- ☐ Imputation fit ONLY on training set
- ☐ Feature selection ONLY on training set
- ☐ Target variable NEVER used in feature engineering
- ☐ Test set NEVER seen during model tuning
- ☐ Outcome 2/3/4 columns DROPPED before any analysis

---

14. CLINICAL CONTEXT & DOMAIN KNOWLEDGE

14.1 What is NOA?

- **Non-Obstructive Azoospermia (NOA):** Complete absence of sperm in the ejaculate due to impaired spermatogenesis (not blockage)
- **Prevalence:** ~1% of all men, ~10% of infertile men
- **Treatment:** Micro-TESE (Microsurgical Testicular Sperm Extraction)
- **Success rate:** ~40-60% depending on etiology (matches our ~37.3% positive class)

14.2 Key Clinical Variables

- **FSH:** Elevated FSH (>12 IU/L) indicates primary testicular failure; very high FSH (>20) = severe
- **Testosterone:** Low (<300 ng/dL) indicates hypogonadism; may benefit from treatment before TESE
- **Testicular Volume:** Smaller testes = less spermatogenic tissue; bilateral atrophy (<4 mL each) = poor prognosis
- **Karyotype:** 46,XY = normal; 47,XXY = Klinefelter syndrome (most common genetic cause)
- **Y Chromosome Microdeletion:** AZFa/b deletions = very poor prognosis; AZFc = some chance of retrieval
- **Pathology (RT/LT):** Histopathological findings from testicular biopsy
- **Sakamoto Score:** Ultrasound-based testicular tissue characterization

14.3 Age Thresholds for NOA

Age Group	Range	Clinical Significance
Young	<35 years	Optimal; baseline reference
Middle-aged	35-39 years	Morphology decline begins
Advanced Paternal Age (APA)	40-44 years	AUA/ASRM counseling threshold
Elderly	≥45 years	Significantly reduced success rates
Binary cutoff	**40 years**	Most common in literature (AUA/ASRM)

14.4 Expected Important Features (from literature)

Based on published NOA prediction studies, these features are typically most important:
1. FSH level (most consistently reported)
2. Testicular volume / size
3. Histopathology pattern
4. Karyotype / genetic status
5. Age
6. Testosterone level
7. Y chromosome microdeletion status
8. LH level
9. Inhibin B (excluded in our dataset — too many missing)
10. BMI (emerging evidence)

14.5 Future Work (to mention in reports)

- External validation on independent cohort
- Temporal validation (train on earlier, test on later patients)
- Web-based CDSS (Clinical Decision Support System) deployment
- Usability testing with physicians
- Cost-effectiveness analysis
- Multi-center validation study
- Prospective validation

APPENDIX A: Feature Mapping JSON Reference

The complete mapping from `Feature_N` → clinical variable name is stored in:

```
[path]
```

This MUST be used in all XAI outputs and clinical reports to replace generic feature names with clinical variable names.

APPENDIX B: Previous Work Status

Synthetic Data Run (Completed — NOT valid for publication)

- Phases 1-5 were previously executed with **synthetic data** generated by `sklearn.make_classification()`
- All results are artificial and must be discarded
- The architecture, code structure, and methodology can be reused
- **AUC scores of 0.98+ were artifacts of synthetic data separability, not real clinical performance**

What Must Be Redone from Scratch

Phase	Must Redo?	Reason
Phase 0	✅ Minor	Replace dataset file only
Phase 1	🔴 FULL REDO	Real data has different patterns, missing values, outliers
Phase 2	🔴 FULL REDO	All statistics are data-dependent
Phase 3	🔴 FULL REDO	Models must be trained on real data
Phase 4	🔴 FULL REDO	Validation depends on trained models
Phase 5	🔴 FULL REDO	XAI depends on trained models

What CAN Be Reused

- ✅ Code structure & methodology
- ✅ Architecture templates
- ✅ Report templates & format
- ✅ Visualization code/style
- ✅ Pipeline design patterns
- ❌ All numerical results
- ❌ All figures & plots
- ❌ Feature rankings

- ❌ Clinical interpretations

APPENDIX C: Key References Library

#	Reference	Used For
1	Chawla NV et al. “SMOTE” JAIR (2002)	Class imbalance - SMOTE
2	Fernández A et al. “Learning from Imbalanced Data Sets” Springer (2018)	Class imbalance - combined approach
3	He H & Garcia EA “Learning from Imbalanced Data” IEEE TKDE (2009)	Class imbalance overview
4	Stekhoven DJ & Bühlmann P. “MissForest” Bioinformatics (2012)	Missing data imputation
5	EMA ICH E9(R1) (2010/2020)	Missing data threshold
6	Collins GS et al. “TRIPOD” BMJ (2015)	Prediction model reporting
7	Guyon I & Elisseeff A. “Feature Selection” JMLR (2003)	Feature selection methodology
8	Aguinis H et al. “Outliers” ORM (2013)	Outlier detection and handling
9	Bouthillier X et al. “Variance in ML Benchmarks” MLSys (2021)	Multiple random seeds
10	Akiba T et al. “Optuna” KDD (2019)	Hyperparameter optimization
11	Bergstra J et al. “Hyper-Parameter Optimization” NIPS (2011)	Bayesian optimization
12	Cohen J “Statistical Power Analysis” (1988)	Power analysis
13	DeLong ER et al. “Comparing AUCs” Biometrics (1988)	AUC comparison
14	Lundberg SM & Lee SI “SHAP” NIPS (2017)	SHAP explanations
15		LIME explanations

#	Reference	Used For
	Ribeiro MT et al. "LIME" KDD (2016)	
16	Saeys Y et al. "Feature Selection in Bioinformatics" Bioinformatics (2007)	Feature selection
17	Vabalas A et al. "ML Validation with Limited Sample" PLoS ONE (2019)	Validation strategy
18	Hastie T et al. "ESL" (2009)	Train/test split
19	James G et al. "ISLR" (2013)	VIF / multicollinearity
20	Potdar K et al. "Categorical Encoding" (2017)	Encoding strategy

---

APPENDIX D: Execution Instructions for New AI agents trained by Hossein Jamalirad Chat

How to Start a New Phase

1. **Upload this document** to the new AI agents trained by Hossein Jamalirad chat
2. **State:** "Execute Phase {N} of the NOA ML Master Specification"
3. **Provide:** The dataset file path (for Phase 0/1) or previous phase outputs
4. **The agent should:**
   - Read this entire specification
   - Follow the exact tasks defined for that phase
   - Use the evidence-based decisions and references specified
   - Generate all required outputs
   - Run the double-check points
   - Update the Decision Logger

Phase Dependencies (Data Flow)

```
Phase 0 → 1_Data/Raw/original_dataset.csv
    ↓
Phase 1 → 1_Data/Cleaned/cleaned_dataset.csv + 1_Data/Processed/encoded_dataset.csv
    ↓
Phase 2 → 3_Results/Phase2_Statistics/ (all stats CSVs)
    ↓
Phase 3 → 6_Models/Saved/*.joblib + 3_Results/Phase3_Models/
    ↓
Phase 4 → 3_Results/Phase4_Validation/ (all validation CSVs)
    ↓
Phase 5 → 5_Reports/ (all final reports + figures)
```

Critical Reminders for Each Phase

- **Always read this specification document first**
- **Always use clinical variable names** (not Feature_N) in reports
- **Always save figures as BOTH PNG and TIFF at 300 DPI**
- **Always update the Decision Logger with references**
- **Always run double-check points before moving to next phase**
- **Never apply SMOTE/scaling/imputation to test data**
- **Never use Outcome 2/3/4 — Outcome 1 only**
- **Target column name:** `Outcome1`
- **Dataset file:** `Book_11.17.2025_Dr. Rad_Final ID.xlsb` → Sheet: `Final Sheet`

End of Master Specification Document
Version 1.0 — Generated 2026-03-14
Project: NOA ML — Non-Obstructive Azoospermia Sperm Retrieval Prediction